

Adaptive Trapdoor Computing: Maintaining Confidentiality Under Distributional Drift

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April 30, 2026

Abstract

TODO. Static confidentiality assumes a fixed query distribution. Real systems see drift. We develop online observation and retuning of the homophonic allocation $K(x)$, prove a regret bound on average confidentiality, and present a secret-preserving retune algorithm that maintains validity of already-emitted cipher values. We evaluate on TBD real drift dataset.

1 Introduction

TODO. See notes/plan.md.

2 Background

TODO. Cite cipher-maps and The Entropy Ratio.

3 Drift and Staleness: Formal Model

TODO.

4 Online Tuning

TODO. Describe observer, allocators, cadence.

5 Secret-Preserving Retune

TODO. Algorithm + correctness theorem.

6 Experiments

TODO. Synthetic + real drift.

7 Discussion

TODO.

8 Related Work

TODO. See `notes/related-work.md`.

9 Conclusion

TODO.

References